

Project Initialization and Planning Phase

Date	30 th June 2026
Team ID	xxxxxxx
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to transform the learning experience by developing an AI-powered educational assistant that provides intelligent question answering, concept explanations, quiz generation, text summarization, and personalized learning recommendations. EduGenie leverages Google's Gemini 1.5 Pro and LaMini-Flan-T5 language models to deliver accurate, interactive, and personalized educational assistance through a user-friendly web interface. The system improves learning efficiency, reduces study time, and enhances students' understanding of complex concepts.

Project Overview	
Objective	The primary objective of EduGenie is to simplify learning by integrating Generative AI and Natural Language Processing technologies into an intelligent educational assistant capable of answering academic questions,
Scope	The project focuses on developing a complete AI-powered educational platform using FastAPI, Google Gemini 1.5 Pro, LaMini-Flan-T5, HTML, CSS, and Jinja2. It includes backend API development, frontend user interface, prompt engineering.
Problem Statement	
Description	Students often struggle to find reliable explanations, create study notes, prepare quizzes, and receive personalized guidance from scattered educational resources. Existing platforms lack an integrated AI-powered solution that provides multiple educational services in one place
Impact	Addressing these challenges will improve learning efficiency, reduce time spent searching for study materials, enhance concept understanding, encourage self-learning.
Proposed Solution	
Approach	Develop an AI-powered educational assistant using FastAPI as the backend framework, integrating Gemini 1.5 Pro for intelligent educational responses and LaMini-Flan-T5 for lightweight text generation tasks.
Key Features	AI-powered educational question answering

	• Educational text summarization • Prompt engineering for accurate educational responses
--	---

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn
Libraries	Additional libraries	Transformers, Torch, Google Generative AI SDK, Jinja2 Visual Studio Code, Python
Development Environment	IDE	3.11+, Git
Data		
Data	Source, size, format	User educational queries, AI-generated responses, quizzes, summaries, and learning recommendations